

<?php

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/* 3 b. Develop a PHP Program to compute the roots of a quadratic
    equation by accepting the coefficients. Print the
    appropriate messages.
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3b_roots.php
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*/
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/* Generic quadratic equation, second-degree polynomial equation
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$$f(x) = y = a x^2 + b x + c = 0$$

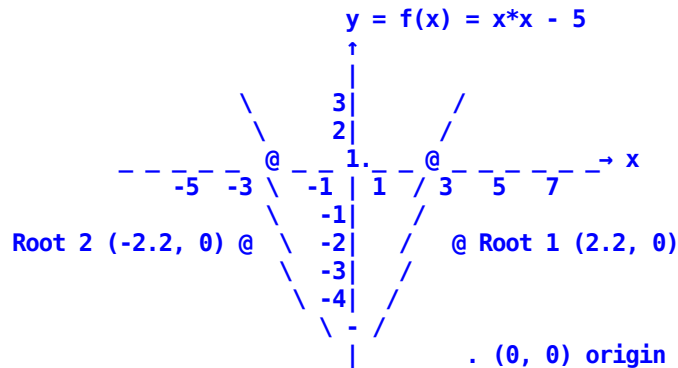
where a and b known as coefficients, and c , all are constants
a $\neq$ 0, else it would be a linear equation, bx + c

Can have up to (atmost) two solutions, known as the roots
values of x, where equation evaluates to zero

graphically, where curve intersects with x axis, y = 0

Example, on x y cartesian coordinate system

$$x^2x - 5 = 0, \text{ as } ax^2 + bx + c = 0, a = 1, b = 0, c = -5$$



Visualize curve, roots on GeoGebra, Graphing Calculator

<https://www.geogebra.org/graphing>

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$$\text{Roots} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{Why/How?}$$

$\sqrt{b^2 - 4ac}$, decides if roots are real ($\mathbb{R}$) or complex ($\mathbb{C}$)

$b^2 - 4ac$, also known as discriminant Δ

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/*
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$$\begin{aligned} \text{numerator} &= -b \pm \sqrt{b^2 - 4ac} \\ \text{denominator} &= 2a \end{aligned}$$

$$\text{Roots} = \text{numerator} / \text{denominator}$$

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*/
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?>

<?php

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$a = 1;
$b = 1; // check with values of b as 1 , or 2 , or 3
$c = 1;

echo "<br /> For a = $a , b = $b , c = $c , roots are : ";
// 2
$disc = $b * $b - 4 * $a * $c; // Δ , discriminant , b - 4ac
$denom = 2 * $a ;

if ( $disc == 0 ) // √ 0 is real
{ // since Δ $disc is 0 , roots = ( -b ± √ 0 ) / 2 a = -b/2a
  $root = -$b / $denom;
  echo "<br /> real , same root = " . $root ;
}
if ( $disc > 0 ) // else if can also be used, in PHP keyword is elseif
{ // √ Δ is real , and two possible values , hence two real roots
  $root1 = ( -$b + sqrt($disc) ) / $denom; //( -b + √ (bb - 4ac) )/2a
  $root2 = ( -$b - sqrt($disc) ) / $denom; //( -b - √ (bb - 4ac) )/2a
  echo "<br /> real, two distinct roots, r1 = $root1 , r2 = $root2";
}
if ( $disc < 0 ) // if elseif else can also be used
{ // Δ is -ve , √ -ve value is complex, represented using i , i = √ -1
  // root has to broken down as real_part + i imaginary_part
  $real_part = -$b / $denom;
  $imag_part = sqrt( - $disc ) / $denom; //√ double negation, √ -(Δ)
  echo "<br /> complex, two distinct roots <br /> ";
  echo "$real_part + i $imag_part, and $real_part - i $imag_part";
} // How could complex roots be displayed graphically
```

?>

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For more on , please check ,

sqrt – Square root

\$ php --rf sqrt

<https://www.php.net/manual/en/function.sqrt.php>

instead of square root function, power function can also be used

pow – Exponential expression

\$ php --rf pow

<https://www.php.net/manual/en/function.pow.php>

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